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Executive Summary

This report serves as the first deliverable of the ENCLOSE project. It provides a comprehensive overview of key European social surveys, examining their complementarities and overlaps. In addition, it introduces two case studies that will be the focus of subsequent deliverables: the first on attitudes towards immigration, based on the integration of the European Social Survey (ESS) Round 10 (2020) and Eurobarometer 93.1 (2020); and the second on the effects of COVID-19 on trust in institutions, analysed using a pseudo-panel derived from two ESS waves.

1. Social Surveys across Europe The report first maps the main surveys relevant for social research in Europe:

- **Surveys by Eurostat:**

- EU Statistics on Income and Living Conditions (EU-SILC) – socioeconomic indicators, poverty, and inequality.
- EU Labour Force Survey (EU-LFS) – employment, unemployment, and labour market dynamics.
- EU Health Interview Survey (EU-HIS) – health status, health determinants, and healthcare use.
- EU Survey on Gender-based Violence – prevalence, attitudes, and reporting of gender-related violence.

- **Surveys by other organisations:**

- Eurobarometer – public opinion on EU institutions, policies, and integration.
- European Social Survey (ESS) – attitudes, beliefs, and behaviour across Europe.
- European Values Study (EVS) – cultural and value orientations.
- European Quality of Life Survey (EQLS) – well-being, work-life balance, and living conditions.
- Survey of Health, Ageing and Retirement in Europe (SHARE) – ageing, health, and socioeconomic conditions of older people.

The report also provides references and links to these data sources.

2. Integrating ESS and Eurobarometer Data A methodological framework is outlined for harmonising ESS and Eurobarometer data:

- **Rationale for selection** — both surveys provide complementary coverage of public attitudes, with different periodicities and topical emphases.
- **Harmonisation process** — alignment of variable definitions and coding schemes to ensure comparability.
- **Case Study 1: Immigration attitudes** — introduced as an application of this framework, based on the integration of ESS Round 10 (2020) and Eurobarometer 93.1 (2020).

3. Integrating ESS Waves A second line of work concerns the harmonisation of two ESS waves:

- **Case Study 2: COVID-19 and institutional trust** — introduced as an investigation of changes in trust in political institutions and satisfaction with national conditions, using a pseudo-panel derived from ESS data.

Main Contributions

- A comprehensive mapping of major European social surveys.
- Discussion of harmonisation strategies for combining different data sources.
- Introduction of two case studies that will be further developed in subsequent deliverables.

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1 Social Surveys across Europe

European social surveys provide invaluable data for examining public attitudes and socio-economic trends across Europe. Their integration within the ENCLOSE project enhances our ability to design evidence-based interventions and policies tailored to the continent's diverse social landscape.

Below we provide a list of various surveys that are interesting from a sociological perspective, emphasising their methods, data accessibility, and contribution to the knowledge of socioeconomic and cultural trends throughout Europe.

1.1 Key Social Surveys conducted by Eurostat

The following sections highlight selected social surveys conducted by Eurostat, focusing on specific topics of interest. The included surveys cover areas such as income and living conditions (EU-SILC), labor market dynamics (EU-LFS), health and healthcare access (EU-HIS), and gender-based violence (EU-GBV). These surveys provide critical insights into various aspects of European societies.

1.1.1 EU STATISTICS ON INCOME AND LIVING CONDITIONS

- **DESCRIPTION:**

The European Union Statistics on Income and Living Conditions (EU-SILC¹) is a harmonised survey conducted annually by Eurostat. The EU statistics on income and living conditions (EU-SILC) aim to collect timely and comparable cross-sectional and longitudinal data on income, poverty, social exclusion, and living conditions. EU-SILC is a household and individual data collection which output is harmonised as it is regulated by legislations.

- **KEY FEATURES:**

- **Research topics:** EU-SILC collects data on various dimensions of socio-economic well-being, including income, poverty, material deprivation, social exclusion, housing conditions, and access to essential services such as healthcare and education.
- **Geographical coverage:** EU-SILC covers all EU member states, as well as some non-EU countries participating in the European Statistical System (ESS) cooperation.
- **Target population:** All private households, and all persons aged 16 and over within the household, are eligible for the operation. Persons living in collective households and in institutions are generally excluded from the target population. Information on social exclusion and housing conditions is collected mainly at the household level. Labour, education, and health information is obtained from individuals aged 16 and over. Income variables at the detailed component level are also mainly collected from individuals.
- **Sampling:** One of the major strength of EU-SILC is the usage of the best sampling frames available in each National Statistical Institute. According to the EU-SILC Framework Regulation, data are to be based on a nationally

¹<https://ec.europa.eu/eurostat/web/income-and-living-conditions/information-data>

representative probability sample of the population residing in private households within the country, irrespective of language, nationality or legal residence status.

- **Data collection:** Eurostat issues yearly methodological guidelines which provide extended explanations and recommendations on the implementation of the data collection.
- **Data accessibility:** Eurostat grants access to microdata for scientific purposes in accordance with Regulation (EU) No 557/2013 on access to confidential data for scientific purposes. To apply for access to Eurostat's microdata, the organisation must first be recognised by Eurostat as a research entity – a university, research institution or research department in public administration, bank, statistical institute etc.
Applying for microdata access²
 1. Step 1 – Apply to have the research organisation recognised as a research entity (the procedure takes around 4 weeks).
 2. Step 2 – Apply for access to microdata. Once Eurostat has recognised the organisation as a research entity, researchers can apply for access by submitting a research proposal (the procedure takes around 8 weeks).

The EU-SILC scientific use files (SUFs) contain partially anonymised microdata for scientific purposes. Along with the removal of the original identifiers in the cross-sectional datasets in accordance with the requests of Member States, methods of SDC have been applied to reduce (in keeping with current best practice) the risk of identification of a statistical unit to an appropriate level.

- **Waves:** Around 90% of the data collection is made up of annual variables. The rest are either modules that are collected every three or six years or modules conducted ad-hoc to reply to policy needs.
- **Cross-sectional and longitudinal design:**
 - * Cross-sectional data (2004-) pertaining to fixed time periods, with variables on income, poverty, social exclusion and living conditions
 - * Longitudinal data (2005-) pertaining to individual-level changes over time, usually observed over four years.

1.1.2 EU LABOUR FORCE SURVEY

• DESCRIPTION:

The European Labour Force Survey (EU-LFS³) is a large-scale survey conducted by Eurostat. The EU-LFS is a large household sample survey providing quarterly results on labour participation of people aged 15 and over and on people outside the labour force. Its primary aim is to gather comprehensive and comparable data on employment, unemployment, and other characteristics of the labor force across European countries.

The main statistical objective of the EU-LFS is to divide the resident population of working age (15 years and above) into three mutually exclusive and exhaustive groups – persons employed, unemployed and outside the labour

²HOW TO APPLY FOR MICRODATA ACCESS?

³<https://ec.europa.eu/eurostat/web/lfs/information-data>

force according to the classification of the International Labour Organisation (ILO) – and to provide descriptive and explanatory data on each of these three groups. Respondents of the EU-LFS are assigned to one of these groups on the basis of the information obtained through the survey questionnaire, which principally relates to their actual activity within a particular reference week.

• **KEY FEATURES:**

- **Research topics:** the EU-LFS collects data on various aspects of the labor market. The core of the EU-LFS data collection covers the following topics: person and household characteristics, labour market participation, educational attainment and background, job tenure, work biography and previous work experience, working conditions including working hours and working time arrangements, participation in education and training, health status and disability, income, consumption and elements of wealth. Moreover, the EU-LFS includes modules on specific topics, among which some are collected with a regular periodicity (of eight years).
- **Geographical coverage:** the EU-LFS is conducted in 34 countries: the EU Member States, three EFTA countries (Iceland, Norway, Switzerland) and four Candidates Countries (Montenegro, North Macedonia, Serbia and Turkey).
- **Target population:** the EU-LFS covers persons aged 15 years and over who live in private households. Those doing military or community service and those living in institutional or collective households are not included.
- **Sampling:** in all 34 participating countries, the EU-LFS is based on a random sampling design. Almost all countries use a stratified random sampling method for the LFS. The stratification is mainly done on geographical areas (NUTS 2 or NUTS 3 or LAU levels). The degree of urbanisation is also a common stratification variable. The most common sample design for the EU-LFS is the stratified two-stage cluster sampling. The achieved sample in the EU-LFS is approximately 0.3% of the total population.
- **Data collection:** The national statistical institutes across Europe conduct the surveys and are responsible for selecting the sample, preparing the questionnaires, conducting the direct interviews among households, and forwarding the results to Eurostat. At European level, data are made comparable across countries and over time by:
 - * using the same concepts and definitions;
 - * following the guidelines of the International Labour Organisation;
 - * using common classifications, such as NACE, ISCO, ISCED, NUTS;
 - * recording the same set of characteristics in each country.
- **Data accessibility:** Eurostat grants access to microdata for scientific purposes in accordance with Regulation (EU) No 557/2013 on access to confidential data for scientific purposes. To apply for access to Eurostat's microdata, the organisation must first be recognised by Eurostat as a research entity – a university, research institution or research department in public administration, bank, statistical institute etc. Applying for microdata access⁴

⁴HOW TO APPLY FOR MICRODATA ACCESS?

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- **Waves:** the EU-LFS provides quarterly and annual data.

1.1.3 EU HEALTH INTERVIEW SURVEY

- **DESCRIPTION:**

The European Health Interview Survey (EHIS⁵) is a large-scale survey conducted by Eurostat. The EHIS aims include assessing the prevalence of health conditions and risk factors, monitoring healthcare access and utilisation patterns, and identifying health inequalities and determinants of health disparities.

- **KEY FEATURES:**

- **Research topics:** the EHIS collects data on a wide range of health-related topics. Specifically EHIS consists of 4 modules on
 - * *background variables* on demography and socio-economic status such as sex, age, education, labour status, etc.
 - * *health status* such as self-perceived health, chronic conditions, limitation in usual activities, disease-specific morbidity, physical and sensory functional limitations, etc.
 - * *health-care use* such as hospitalisation, consultations, unmet needs, use of medicines, preventive actions, etc.
 - * *health determinants* such as height and weight, fruit and vegetable consumption, smoking, alcohol consumption, etc.
- **Geographical coverage:**
 - * first wave: 17 EU countries – Austria, Estonia, Slovenia, Belgium, Bulgaria, Czechia, Cyprus, France, Latvia, Malta, Romania, Greece, Spain, Hungary, Poland, and Slovakia, Germany;
 - * second wave: all EU countries, Iceland, and Norway;
 - * third wave: all EU countries, Iceland, Norway, Serbia, Albania, and Turkey
- **Target population:** persons aged 15 and over usually residing in private households in the territory of the Member State concerned at the time of the data collection.
- **Sampling:** representative samples drawn from sampling frames set up at national level.
- **Data collection:** data in the EHIS are collected using standardized questionnaires developed by Eurostat in collaboration with national statistical offices and health authorities. Personal (face-to-face) interviews are preferred mode of data collection to be used and the model questionnaire refers to this mode. If other modes of data collection are used the questionnaire should be adapted accordingly to better serve the

⁵<https://ec.europa.eu/eurostat/en/web/products-manuals-and-guidelines/w/ks-gq-24-018>

respective type of interview. Detailed reporting on the use of modes of data collection is recommended to enable an analysis of possible impacts of different modes on results.

- **Data accessibility:** Anonymised microdata are available for all participating countries, except for France, Albania, and Turkey.
- **Waves:** Until 2019, the survey was conducted every 5 years. As of 2019 onwards, the survey has a 6-year periodicity. The first wave of the survey (EHIS 1) took place between 2006 and 2009. The second wave (EHIS 2) took place between 2013 and 2015. The third wave (EHIS 3) was conducted in 2019.

1.1.4 EU SURVEY ON GENDER-BASED VIOLENCE

• DESCRIPTION:

The aim of the EU survey on Gender-Based Violence (EU-GBV)⁶ survey is to measure gender-based violence as defined in the Istanbul Convention. To collect the data according to this definition, the survey focuses on different types of violence experienced and the relationship between victim and perpetrator. Thus, the survey also includes other forms of inter-personal violence, in addition to gender-based violence against women.

• KEY FEATURES:

- **Research topics:** the EU-GBV survey covers psychological, physical and sexual violence by intimate partner, physical and sexual violence by non-partner, sexual harassment at work, violence experienced in childhood and stalking by any perpetrator.
 - * *Psychological violence* by intimate partner includes a range of behaviours, encompassing acts of emotional abuse and controlling behaviour towards the respondent: belittling and humiliating; forbidding the respondent to see friends or family, or to engage in hobbies or other activities; tracking the respondent via GPS, phone or social network; forbidding the respondent to leave the house without permission or locking the respondent up; constantly accusing the respondent of being unfaithful or getting angry if the respondent speaks to another person; forbidding the respondent to work; controlling the finances of the whole family and the respondent's personal expenses; keeping or taking away the respondent's ID card/passport to control the respondent; yelling and smashing things or behaving in a certain way with the aim of scaring or intimidating the respondent; threatening to hurt the respondent's children or other people close to the respondent; threatening to take away the respondent's children or to deny custody; and threatening to harm them self if the respondent leaves them.
 - * *Physical violence* refers to a range of violent types of behaviour or acts involving harm and fear, such as threatening to harm the respondent; pushing or shoving the respondent, pulling their hair, slapping or throwing something at them; punching the respondent or beating them with an object; kicking; burning (with fire or acid or by some other

⁶<https://ec.europa.eu/eurostat/web/products-manuals-and-guidelines/-/ks-gq-21-009>

means); trying to choke or strangle the respondent; threatening to use or actually using a knife, gun, acid or something similar; or using force against the respondent in some other way with the aim of hurting them.

- * *Sexual violence* includes unwanted sexual intercourse through force or physical violence or by exploiting a situation in which the respondent is not able to refuse sexual intercourse because they are under the influence of alcohol or drugs. It also includes unwanted sexual intercourse which the respondent is too afraid to refuse and cases in which the respondent is forced into unwanted sexual intercourse with another person or persons. Attempts to carry out any of the above acts or any other unwanted sexual behaviour that the respondent finds degrading or humiliating are included as well. Finally, this type of violence also covers unwanted sexual touching by non-partners. Sexual violence in the childhood covers experiences before the age of 15 perpetrated by any person, such as posing naked in front of another person, unwanted sexual touching or sexual intercourse. Type of perpetrators are defined as for non-partner violence.
- * *Sexual harassment at work* covers the following unwanted behaviours with sexual connotations that happen in a work context: inappropriate staring or leering; being exposed to sexually explicit images or videos; indecent sexual jokes or offensive remarks about a person's body or private life; inappropriate invitations to go out on a date or suggestions for sexual activity of any kind; unsolicited physical contact; inappropriate advances on social networking websites or sexually explicit emails or text messages; threatening with unpleasant consequences if sexual proposals or advances are refused; and any other similar behaviour with sexual connotations that took place at work or work-related settings and that offended, humiliated, or intimidated the respondent.
- * *Stalking* includes a range of offensive or threatening forms of behaviour or acts repeated in the course of the respondent's life: sending unwanted messages, including messages on social media, emails and letters, or gifts; making obscene, threatening, nuisance or silent telephone calls; trying persistently to contact with the respondent, waiting or loitering outside the respondent's home, school or workplace; following or spying on the respondent in person; intentionally damaging the respondent's things (car, motorbike, letterbox, etc.) or the belongings of people close to the respondent, or harming animals belonging to the respondent; making offensive or embarrassing comments about the respondent in public, including comments on social networks; publishing photos, videos or highly personal information about the respondent.
- **Geographical coverage:** 18 Member States implement the EU-GBV survey nationally (BE, BG, DK, EE, EL, ES, FR, HR, LV, LT, MT, NL, AT, PL, PT, SK, SI, FI). Italy agreed to share the data based on its national Violence Against Women survey. For the remaining eight Member States (CZ, DE, IE, CY, LU, HU, RO, SE), the European Union Agency for Fundamental Rights (FRA) and the European Institute for Gender Equality (EIGE) implemented a separate data collection following the Eurostat methodological manual. Concerning the survey implementation in other countries, Iceland implements the survey nationally while,

under the Instrument for Pre-Accession Assistance (IPA) as part of the multi-beneficiary statistical cooperation, four countries (ME, MK, RS and XK) implement the EU-GBV survey and two other countries (BA and AL) conducted some methodological work.

- **Target population:** persons aged 18-74 who live in private households, with a focus on women. However, men can be included in the target population by countries willing to do so. As a result, around half of countries included men, but indicators are calculated and disseminated only for women.
- **Sampling:** data are based on a nationally representative sample of the population residing in private households within the country, irrespective of language, nationality or legal residence status. The sampling frame and methods of sample selection are left to the countries' experiences with other population surveys in which they could ensure that every individual in the target population is assigned a known and non-zero probability of selection. The main condition that needs to be fulfilled is that only one person can be interviewed per household. The indicative effective sample size defined for the EU-GBV is 5000 respondents per country. However, countries are allowed to increase or decrease the sample size in order to collect high quality data.
- **Data collection:** mode of data collection can vary from country to country. However, due to the sensitive nature of this survey, and research and data-based findings, Eurostat recommends face-to-face and computer-assisted methods. The possibility of self-completion is also supported.
- **Data accessibility:** EU-GBV survey scientific use files contain anonymised microdata. To appropriately reduce the risk of disclosure of respondents or information about them, the anonymisation process focusses on the background variables, as the survey collects information about respondents' personal experiences that are neither available in registers nor publicly.

Eurostat grants access to microdata for scientific purposes in accordance with Regulation (EU) No 557/2013 on access to confidential data for scientific purposes. To apply for access to Eurostat's microdata, the organisation must first be recognised by Eurostat as a research entity – a university, research institution or research department in public administration, bank, statistical institute etc.

Applying for microdata access⁷

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 2. Step 2 – Apply for access to microdata. Once Eurostat has recognised the organisation as a research entity, researchers can apply for access by submitting a research proposal (the procedure takes around 8 weeks).
- **Waves:** data collection for the first wave covers the 2020-2024 period.

⁷HOW TO APPLY FOR MICRODATA ACCESS?

1.2 Key Social Surveys conducted by Other Organisations

The following sections present major social surveys carried out by organisations other than Eurostat. These include the Eurobarometer, the European Social Survey (ESS), the European Values Survey (EVS), the European Quality of Life Survey (EQLS), and the Survey of Health, Ageing, and Retirement in Europe (SHARE), each offering valuable data on various dimensions of societal trends, attitudes, and behaviors across Europe.

1.2.1 EUROBAROMETER

- **DESCRIPTION:**

Eurobarometer is a series of public opinion surveys conducted regularly across EU member states, and occasionally in candidate and potential candidate countries, on behalf of EU institutions. Established in 1974, it serves as the EU's official polling tool, used by the European Parliament, European Commission, and other agencies to monitor public opinion on EU-related matters and broader political or social issues.

In 2022, the European Parliament and the European Commission have jointly launched a new inter-institutional Eurobarometer website ⁸. This platform hosts in one place all existing Eurobarometer survey data from both the European Parliament and the European Commission, reaching back to 1974. The website offers full access to the largest single survey database in the European Union.

The Eurobarometer is structured into distinct survey types, each designed to address specific research needs for the European Union (EU) institutions. These types include Standard Eurobarometer, Special Eurobarometer, Flash Eurobarometer, Qualitative Studies.

Special Eurobarometer surveys are in-depth thematic studies relevant to the activities of the European institutions. These can range, for example, from topics such as climate change to social issues, cybersecurity issues or discrimination.

Flash Eurobarometer surveys are ad-hoc thematic surveys, carried out within a short timespan on a wide variety of specific topics relevant to the activities of the European institutions. They are characterised by reliance on either telephone or online interviewing methods. Flash Eurobarometer surveys are mostly conducted in the EU Member States.

Qualitative studies assess in detail the attitudes, opinions, expectations, concerns etc. of the public or of selected social groups through focus groups or in-depth individual interviews.

- **KEY FEATURES of the Standard Eurobarometer:**

- **Research topics:** the standard modules ask for attitudes towards European unification, institutions and policies, complemented by measurements for general socio-political orientations, as well as by respondent and household demographics.

⁸<https://europa.eu/eurobarometer/screen/home>

- **Geographical coverage:** initially the Eurobarometer surveys were only conducted in EU/EC member countries and in accordance with the enlargement process. In addition and starting with Eurobarometer 62 (2004) Accession (AC) and Candidate Countries (CC) are included intermittently. Occasionally one or more EFTA countries are also surveyed. The United Kingdom is being surveyed in the framework of the Standard Eurobarometer surveys since its withdrawal from the European Union.
- **Target population:** persons of any nationality of an European Union member country, aged 15 years and over, resident in any of the Member States.
- **Sampling:** the basic sampling design in all member states is a multi-stage, random (probability) one. The sampling is based on a random selection of sampling points after stratification by the distribution of the national, resident population in terms of metropolitan, urban and rural areas, i.e. proportional to the population size (for a total coverage of the country) and to the population density. These primary sampling units (PSU) are selected from each of the administrative regions in every country. In the second stage, a cluster of addresses is selected from each sampled PSU. Addresses are chosen systematically using standard random route procedures, beginning with an initial address selected at random. In each household, a respondent is selected by a random procedure, such as the first birthday method.
- **Data collection:** interviews are conducted face-to-face (CAPI) in respondents' homes.
- **Data accessibility:** the primary data of the Eurobarometer series on microdata level and the related documentation are placed at the disposal of the scientific community. They are curated at the GESIS Survey Data Curation department ⁹ and at the Interuniversity Consortium for Political and Social Research (ICPSR)¹⁰.
- **Waves:** conducted twice a year (spring and autumn).
- **Cross-sectional design:** for each Standard Eurobarometer survey new and independent samples are drawn.

1.2.2 EUROPEAN SOCIAL SURVEY

• DESCRIPTION:

The European Social Survey (ESS)¹¹ is an academically driven cross-national survey that has been conducted across Europe since its establishment in 2001. It aims to measure the attitudes, beliefs, and behavior patterns of individuals across Europe. Conducted every two years, it covers a wide range of topics related to social and political life.

• KEY FEATURES:

- **Research topics:** the main objective of the European Social Survey is to provide high-quality data on social attitudes, values, and behavior in Europe. The topics covered at least once by the ESS since its

⁹<https://www.gesis.org/en/eurobarometer-data-service/data-and-documentation>

¹⁰<https://www.icpsr.umich.edu/web/ICPSR/series/00026>

¹¹<https://www.europeansocialsurvey.org/>

inception include media and social trust, politics, subjective well-being, gender, household, socio demographics, human values, immigration, citizen involvement, health and care, economic morality, family, work and well-being, timing of life, personal and social well-being, welfare attitudes, ageism, trust in justice, democracy, health inequalities, climate change and energy use, justice and fairness, and digital social contacts.

The ESS questionnaire uses a combination of repeated key items (the *core section*) and rotating modules. The core allows change and continuity across a wide range of social variables to be monitored. The inclusion of rotating modules increases the scope of the survey as they focus on a specific topic in depth. In addition to the core and rotating sections of the questionnaire, a third section contains (1) the well-established 21-item measure of human values developed by the Israeli psychologist Professor Shalom Schwartz and used to classify respondents according to their basic value orientations, and (2) a number of repeated measures from the main questionnaire asked in a slightly modified form, using the Multi-Trait Multi-Method (MTMM) approach to evaluate the reliability and validity of the items, in an effort to measure and improve the quality of the questions fielded.

- **Geographical Coverage:** the ESS covers a large number of European countries, including EU member states, candidate countries, and other European countries. The number of participating countries has varied across survey rounds, with over 30 countries participating in recent waves.
- **Target population:** people aged 15+ living within a country.
- **Sampling design:** the ESS uses a multistage, probability sampling method to ensure that the samples are representative of the population in each country. Each country is required to achieve an effective sample size of 1,500 completed interviews, taking into account potential design effects due to the clustering of the sample and/or the variation in inclusion probabilities. The data is weighted to ensure that the sample of respondents accurately reflects the demographics of the entire population.
- **Data collection:** data has mainly been collected via face-to-face CAPI interviews in all participating countries. It was announced in May 2022 that the ESS General Assembly has endorsed a recommendation from its Core Scientific Team to plan a transition of its data collection. This will see a switch from face-to-face interviews to a (web first) self-completion design. The design will involve web and paper self-completion surveys.
- **Data accessibility:** the European Social Survey provides free and open access to its data for research purposes. Data from all survey rounds, along with detailed documentation and analysis tools, are available on the ESS website¹².
- **Waves:** European Social Survey (ESS) had conducted 13 rounds of data collection.
- **Cross-sectional design:** every two years across Europe since 2002.

¹²<https://ess.sikt.no/en/?tab=overview>

1.2.3 EUROPEAN VALUES STUDY

- **DESCRIPTION:**

The European Values Study (EVS)¹³ is a large-scale, cross-national, and longitudinal survey research program on basic human values. It provides insights into the ideas, beliefs, preferences, attitudes, values, and opinions of citizens all over Europe. It is a unique research project on how Europeans think about life, family, work, religion, politics, and society. The EVS is conducted approximately every nine years.

- **KEY FEATURES:**

- **Research topics:** the main objective of the EVS is to investigate the diversity of human values and beliefs (life, family, work, religion, politics, society) across European countries and to understand how these values change over time. The survey seeks to explore the cultural and social factors that shape individuals' attitudes and behaviors.

The EVS questionnaire covers a wide range of topics related to values, beliefs, attitudes, and behaviors. Some of the key themes addressed in the survey include religion, family, politics, morality, gender roles, and social trust. The questionnaire is designed to be comparable across countries and survey waves to facilitate cross-national analysis.

- **Geographical coverage:** the surveys have been conducted in a nine-year cycle since 1981. The EVS covers a wide range of European countries, including EU member states, candidate countries, and other European countries. The number of participating countries has varied across survey waves, with over 40 countries participating in recent waves.
- **Target population:** resident population older than 18 years old
- **Sampling:** the use of quotas was admitted in the first waves, but since 2008 only probabilistic representative sample.
- **Data collection:** data in the EVS is collected through face-to-face interviews conducted by trained interviewers.
- **Data accessibility:** data and documentation of all the waves are available free of charge¹⁴. They are stored at GESIS – Leibniz Institute for the Social Sciences in Cologne, Germany.
- **Waves:** five waves of data collection. It was first conducted in 1981 and then repeated in 1990, 1999, 2008, and 2017.
- **Longitudinal design:** the EVS follows a longitudinal design, collecting data from the same individuals over multiple survey rounds.

1.2.4 EUROPEAN QUALITY OF LIFE SURVEY

- **DESCRIPTION:**

The European Quality of Life Survey (EQLS)¹⁵ is a multidisciplinary survey conducted by the European Foundation for the Improvement of Living and Working Conditions (Eurofound). The survey aims to provide comprehensive insights into the quality of life, well-being, and living conditions of individuals

¹³<https://europeanvaluesstudy.eu/>

¹⁴<https://europeanvaluesstudy.eu/methodology-data-documentation/data-and-documentation/>

¹⁵<https://www.eurofound.europa.eu/en/surveys/european-quality-life-surveys-eqls>

across European countries. Since its launch in 2003, the EQLS has developed into a valuable set of indicators which complements traditional indicators of economic growth and living standards such as GDP or income. The EQLS indicators are as clear and appealing as GDP, however they are more inclusive of environmental and social aspects of progress and therefore are easily integrated into the decision-making process and taken up by public debate at EU and national levels in the European Union. The EQLS examines both the objective circumstances of European citizens' lives and how they feel about those circumstances and their lives in general. The main objectives of the survey are:

- identifying groups at risk and issues of concern as well as of progress;
- monitoring trends by providing homogeneous indicators on these issues;
- contributing to European policy development on living conditions.

• **KEY FEATURES:**

- **Research topics:** the primary objective of the EQLS is to assess and monitor the quality of life and well-being of individuals living in European countries. The survey examines various dimensions of quality of life, including material living conditions, social relationships, health, subjective well-being, and satisfaction with life.
The EQLS questionnaire covers topics such as income and material deprivation, employment status, housing conditions, social support networks, health and disability, and overall life satisfaction.
- **Geographical coverage:** the EQLS covers a wide range of European countries, including EU member states, candidate countries, and other European countries. While the number of participating countries may vary across survey rounds, the survey typically includes data from a significant number of countries to ensure broad representation.
- **Target population:** adult population (18+) living in private households.
- **Samplig:** the EQLS uses a rigorous sampling methodology to ensure that the samples are representative of the population in each country.
- **Data collection:** data in the EQLS is collected through face-to-face interviews conducted by trained interviewers.
- **Data accessibility:** Eurofound's survey datasets are made available no later than two years after fieldwork completion. The Eurofound datasets are stored with the UK Data Service (UKDS) in Essex, UK. The data are available free of charge to those who intend to use them for non-commercial purposes. Requests for use for commercial purposes will be forwarded to Eurofound for authorisation.
- **Waves:** four waves (2003-2007-2012-2016).
- **Cross-sectional design.**

1.2.5 SURVEY OF HEALTH, AGEING AND RETIREMENT IN EUROPE

• **DESCRIPTION:**

The Survey of Health, Ageing and Retirement in Europe (SHARE)¹⁶ is a

¹⁶<https://share-eric.eu/>

comprehensive and longitudinal study that aims to understand the dynamics of aging populations in Europe. It collects data on various aspects of health, economic status, social networks, and well-being among older adults aged 50 and over. SHARE provides valuable insights into the challenges and opportunities associated with population aging, informing policies and interventions to promote healthy and active aging across European countries.

• **KEY FEATURE:**

- **Research topics:** SHARE collects data on various research topics related to aging and well-being, including:
 - * Health and functional status: assessing physical health, cognitive function, and activities of daily living.
 - * Economic status: examining income, wealth, employment status, pension arrangements, and financial security.
 - * Social networks: Investigating social relationships, support networks, and social participation.
 - * Life satisfaction and well-being: measuring subjective well-being, quality of life, and satisfaction with different domains of life.
- **Geographical coverage:** SHARE covers a wide range of European countries, including EU member states, candidate countries, and other European countries. The geographical diversity of SHARE enables researchers to examine variations in aging trajectories, health outcomes, and social determinants across different regions and contexts.
- **Target population:** all persons aged 50 years and over at the time of sampling who have their regular domicile in the respective SHARE country.
- **Sampling:** all countries included in SHARE would have a probability-based sample based on an official person register covering the population of interest. The most frequently used sampling design in the SHARE countries is a multi-stage stratified sampling design, i.e. the country is divided into several strata in a first step to ensure representativeness of different geographical areas within the country, to improve efficiency of the survey estimates, and to reduce the costs of the interview process. Within these strata, PSUs are usually drawn in a second step – often with a probability proportional to their size to give larger PSUs a larger probability of being sampled. Individuals or households/addresses can be drawn within the selected PSUs depending on the available information. All SHARE respondents who were interviewed in any previous wave are part of the longitudinal sample. Additionally, refreshment samples are drawn regularly to maintain representation of the younger age-cohorts of the target population that were not age-eligible in previous waves and compensate for the reduction in panel sample size due to attrition.
- **Data collection:** SHARE collects data through face-to-face interviews (computer-assisted personal interviewing – CAPI) conducted by trained interviewers. The survey employs standardized questionnaires to ensure comparability of data across countries and waves. Data collection methods may vary slightly across countries to accommodate cultural and linguistic differences.
- **Data accessibility:** access to the data collected and generated in the SHARE projects is provided free of charge for scientific use globally, subject

to European Union and national data protection laws as well as the publicly available Conditions of Use. The SHARE data are distributed by SHARE-ERIC (Survey of Health, Ageing and Retirement in Europe – European Research Infrastructure Consortium) to registered users through the SHARE Research Data Center¹⁷.

- **Waves:** SHARE has had nine waves of data collection to date.
- **Longitudinal design:** SHARE follows a longitudinal design, meaning it collects data from the same individuals over multiple waves of data collection. This allows researchers to track changes in health, economic circumstances, and social relationships over time, providing a dynamic picture of aging processes in Europe.

1.3 Links to Social Survey Data Sources

This section provides links to resources containing microdata and metadata for the social surveys discussed in this report, categorised by Eurostat surveys and those conducted by other organizations.

• EUROSTAT

- Microdata
- Datasets availability
- EU-SILC microdata
- EU- SILC metadata
- EU-LFS microdata
- EU-LFS metadata
- EHIS microdata
- EHIS metadata
- EU-GBV microdata
- EU-GBV metadata

• OTHER ORGANISATIONS

- Eurobarometer
- Eurobarometer data and documentation
- ESS: European Social Survey
- ESS data portal
- EVS: European Values Study
- EVS data and documentation
- EQLS: European Quality Of Life Survey
- EQLS data availability
- SHARE: Survey of Health, Ageing and Retirement in Europe
- SHARE data access

¹⁷<https://share-eric.eu/data/data-access>

2 Integrating ESS and Eurobarometer Data

One of the objectives of the ENCLOSE project is to gain a deeper understanding of social phenomena where the information is not jointly observed in a single survey, by applying advanced statistical matching techniques. These methods facilitate the seamless integration of diverse datasets, enabling more accurate and comprehensive analyses of social attitudes and behaviours. Specifically, by combining data from different surveys, the ENCLOSE project aims to generate valuable insights that will contribute to the development of an integrated dataset. In particular, we will focus on integrating two key European surveys: the European Social Survey (ESS) Round 10 (2020) and Eurobarometer 93.1 (2020).

The integration of the ESS and Eurobarometer datasets will follow a statistical matching approach in line with the ENCLOSE project's objectives. The key steps in the methodology are as follows:

1. *Data Harmonisation*: ensuring that common variables are coded consistently across both datasets. This step is critical for ensuring comparability between the two datasets.
2. *Statistical Matching*: employing methods SM techniques to combine data and create a dataset that accurately reflect social attitudes.
3. *Comparative Analysis*: conducting analyses that examine target variables across different demographic groups, countries, and regions. This will highlight variations in attitudes towards immigration, allowing for a deeper understanding of the factors influencing these attitudes.

In this report we consider the first step related to data harmonisation.

2.1 Rationale for data source selection

In line with the ENCLOSE project's aim of utilizing diverse social survey datasets, the European Social Survey (ESS) and Eurobarometer surveys were selected based on their rigorous methodologies, robust sampling designs, and significant relevance to understand social aspects within the EU. Both surveys have long-established reputations for producing reliable data that provide insights into a wide range of social issues, including immigration, political engagement, and public opinion.

The ESS, launched in 2001, offers high-quality data on various European attitudes, beliefs, and behaviours, making it particularly relevant for the ENCLOSE project. A distinguishing feature of the ESS is its standardized methodology, which ensures consistency and comparability across different countries and multiple survey rounds. This rigorous approach enables researchers to conduct both cross-national analyses and to investigate temporal trends on macro parameters. Such consistency is crucial for comparing attitudes across the European countries and identifying shifts in public opinion.

The Eurobarometer surveys, which date back to the 1970s, have also played a pivotal role in capturing the diverse perspectives of EU citizens. The ENCLOSE project specifically utilises Eurobarometer 93.1 (2020) because of its suitability for policy-focused data integration. This survey was designed to assess public opinion

on a range of issues relevant to the EU. The insights derived from this survey are directly applicable to policy discussions and decisions within the EU.

Eurobarometer 93.1, conducted in 2020, provides a parallel timeframe for comparison with ESS Round 10, enhancing the robustness of cross-survey analyses. Additionally, Eurobarometer 93.1 offers extensive European coverage, making it ideal for conducting cross-country comparisons and providing a comprehensive understanding of public sentiment across the EU.

Both surveys use robust random sampling techniques to ensure that the samples are representative of the population at country level. This high-quality sampling is essential for integrating the data meaningfully, enhancing the reliability and validity of the findings. A representative sample allows for accurate generalisations about public attitudes, which is particularly important for informing effective policy-making and social research.

In addition, both the European Social Survey (ESS) and the Eurobarometer surveys provide free access to their microdata for scientific purposes upon registration. They offer comprehensive documentation, including detailed methodologies, standardized questionnaires, and robust codebooks for variables. The datasets, available in SPSS format and fully compatible with R, retain all variable and label specifications, making them highly suitable for advanced analyses. These features are particularly valuable for data integration procedures, such as statistical matching, where harmonised and well-documented data are essential to ensure consistency, comparability, and reliability when combining insights from both sources.

2.2 Harmonisation and reconciliation of multiple sources

The first step in a data matching process is the harmonisation of multiple sources.

When two sources A and B have not been planned to be homogeneous, different actions should be carried out.

Van der Laan (2000) provided the first model for a micro-integration process:

- a) harmonisation of the definition of units: *are the statistical units defined uniformly in all sources?*
- b) harmonisation of reference periods: *do all data refer to the same period or the same point in time?*
- c) completion of populations: *do all sources cover the same target population?*
- d) harmonisation of common variables: *are corresponding variables defined in the same way?*
- e) harmonisation of classifications: *are corresponding variables classified in the same way?* – some common variables need to be recoded and recategorised;
- f) adjustment for measurement errors;
- g) adjustment for missing data;
- h) derivation of variables – some variables are substituted by new variables, by transformation of the available information.

D'Orazio et al. (2006) point out that actions (a)–(e) are characteristic of the harmonisation phase and tend to be fairly ad hoc. There is no universally optimal procedure for the harmonisation phase. In fact, it often involves simplifying the key characteristics of the surveys. In the harmonisation of variables (d), if information in the original sources differ, the information from both the sources is transposed to the same concept. If sources contain different classifications (e) of the variables, these should be converted to a common classification. A specific part of this is the use of standard classifications (Bakker, 2011).

Actions (f)–(g) can rely on an extensive body of statistical literature and are essential when nonsampling errors influence A and B.

Finally, action (h) is undertaken to derive new variables from those observed in A and B.

The empirical studies done in Eurostat showed that in an applied context these standardisation issues can hamper the successful application of matching methods (Leulescu & Agafitei, 2013).

2.2.1 Ensuring consistency between common variables

The quality of results from matching procedures depends on having a common set of variables that are statistically consistent across datasets. For effective matching, the two sample surveys (A and B) should estimate the same distribution for each common variable, representing the same population. Matching variables should exhibit similar joint and marginal distributions in both datasets. Leulescu and Agafitei (2013) discuss different techniques to assess those similarities.

A simple method to assess similarity is by comparing the weighted frequency distributions of each variable in both datasets, with significant differences indicating a lack of coherence. However, this method lacks theoretical grounding, and the threshold for acceptable differences is arbitrary (Leulescu & Agafitei, 2013).

A more refined approach uses distance metrics, like the Hellinger distance, which quantifies the similarity between probability distributions. The Hellinger distance, suggested by Leulescu and Agafitei (2013) for assessing the similarity in the distribution of the common variables, ranges from 0 (perfect similarity) to 1 (total discrepancy), with values below 0.05 often considered indicative of close distributions. Although useful, it does not account for sampling variability or the complexity of categorical data, and its thresholds are arbitrary.

Statistical tests such as Chi-square, Kolmogorov-Smirnov, and Rao-Scott tests offer a more robust framework by considering complex sampling designs, but their application is limited when sampling design information is unavailable.

In cases of substantial differences in distributions, harmonisation procedures like variable recategorisation or calibration techniques may be necessary to improve alignment. However, these methods may not always yield satisfactory results, and in some cases, no harmonisation technique will sufficiently align the datasets.

2.3 First case study: Investigating attitudes towards immigration across Europe

In our first case study, we will focus on exploring attitudes towards immigration across Europe.

Both the ESS Round 10 (hereafter ESS10) and Eurobarometer 93.1 offer resources for understanding the complexities of public opinion on immigration within the broader context of European challenges.

- **ESS10**

The investigation of attitudes towards migration is carried on through a dedicated set of questions included since first ESS edition. This fixed module focuses on the opinion and behaviour of individuals towards immigrants, the assessment of the consequences of immigration, the perception of belonging to a discriminated group, and the determinants of attitudes towards immigrants.

- **Eurobarometer93.1:**

The standard modules explore attitudes towards European unification, institutions, and policies. Within this framework, attention is given also to public opinion on a common European migration policy, investigating citizens' stances on its desirability. This module provides valuable insights into how Europeans perceive coordinated migration strategies at the EU level and their implications for member states.

2.3.1 Geographical coverage and target population

- **ESS10**

- **Geographical coverage:** a total of 31 countries participated in ESS10, including 23 European Union member states (Austria, Belgium, Bulgaria, Cyprus, Croatia, Estonia, Finland, France, Germany, Greece, Ireland, Italy, Latvia, Lithuania, Netherlands, Poland, Portugal, Czech Republic, Slovakia, Slovenia, Spain, Sweden, and Hungary) and 8 non-EU countries (Iceland, Israel, North Macedonia, Montenegro, Norway, United Kingdom, Serbia, and Switzerland).
- **Target population:** people aged 15+ living within a country.

- **Eurobarometer93.1**

- **Geographical coverage:** the standard Europe-wide survey was carried out in 34 countries/territories (the EU, the UK, 5 candidates for EU membership (North Macedonia, Turkey, Montenegro, Serbia and Albania) and the Turkish Cypriot community (in the part of the country not controlled by the Cypriot government)).
- **Target population:** persons of any nationality of an European Union member country, aged 15 years and over, resident in any of the Member States.

2.3.2 Target variables

In our analysis, we aim to investigate how predispositions towards immigration influence the stance of European citizens on a common European immigration policy. Information regarding citizens' stances on this issue is primarily drawn from the Eurobarometer Standard Survey, where respondents are asked to express whether they are in favour of or against a common EU policy on migration, a common EU asylum system, and the reinforcement of EU external borders.

While the Eurobarometer provides valuable insights into general attitudes toward immigration, its questions tend to remain broad and lack detailed exploration of citizens' views.

In contrast, the European Social Survey (ESS) offers more detailed and nuanced questions that allow for a deeper understanding of attitudes toward immigration and related policies.

Therefore, to achieve a more comprehensive analysis, we have decided to integrate data from both the Eurobarometer and the ESS. In the following sections, we will present the target variables in detail, outlining how they are derived and applied to our investigation.

• ESS10

In our study we will consider as target variables two latent traits: *"Acceptability of Immigration by Ethnic and Economic Affinity"* and *"Perceptions of Immigration's Societal Contributions"*.

Y_1 : "Acceptability of Immigration by Ethnic and Economic Affinity"
measured by the following questions:

- "To what extent do you think [our country] should allow people of the same race or ethnic group as most [country] people to come and live here?" (imsmetn).
- "To what extent should [our country] allow people of a different race or ethnic group from most [country] people to come and live here?" (imdfetn).
- "To what extent should [our country] allow people from poorer countries outside Europe to come and live here?" (impcntr).

Responses to all three items are categorised as: 1: "Allow many", 2: "Allow some", 3: "Allow a few", 4: "Allow none".

Y_2 : "Perceptions of Immigration's Societal Contributions"
measured by the following questions:

- "Would you say that people who come to live here generally make the country's economy better or worse?" (imbgeco).
- "Would you say that people who come to live here make the country's cultural life richer or poorer?" (imueclt).
- "Would you say that people who come to live here make [our country] a better or worse place to live?" (imwbcnt).

Responses to all three items are on a scale from 0 to 10, where: 0: "Extremely negative" and 10: "Extremely positive".

• Eurobarometer93.1:

In our study, we will consider as the target variable a latent trait measuring the opinion on whether the respondent is in favour of or against a European Union's common policy on asylum, migration, and border control.

Z: "European Integration in Migration and Border Management"
measured by the following question

- "What is your opinion on each of the following statements? Please tell me for each statement, whether you are for it or against it."
 - * A common European policy on migration (qb5_4).
 - * A common European Asylum system (qb6_1).
 - * A reinforcement of EU external borders with more European border guards and coast guards (qb6_2).

Responses are coded as: 1: "For" 2: "Against".

Table 1 provides a summary of the target variables, while Figure 1 offers a graphical representation of the hypothesised relationships between the latent variables and their corresponding observed indicators.

Latent Trait	Variable	Code
ESS10		
<i>Y</i> ₁ : Acceptability of Immigration by Ethnic and Economic Affinity	Allow many/few immigrants of same race/ethnic group as majority	imsmetrn
	Allow many/few immigrants of different race/ethnic group from majority	imdfetrn
	Allow many/few immigrants from poorer countries outside Europe	impcntr
<i>Y</i> ₂ : Perceptions of Immigration's Societal Contributions	Immigration bad or good for country's economy	imbgeco
	Country's cultural life undermined or enriched by immigrants	imueclt
	Immigrants make country worse or better place to live	imwbcnt
Eurobarometer93.1		
<i>Z</i> : European Integration in Migration and Border Management	EU proposals: common migration policy	qb5_4
	EU proposals: common asylum system	qb6_1
	EU proposals: reinforce external borders with more guards	qb6_2

Table 1: Target variables

To assess the quality of the matching procedures, considering the Eurobarometer dataset as the receiver, we will evaluate the correlation of the

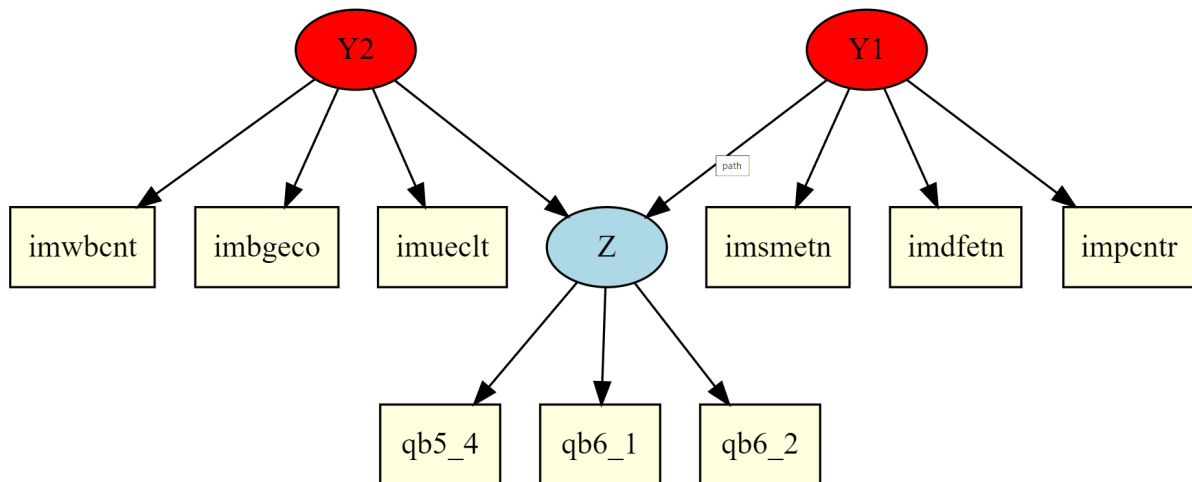


Figure 1: Path diagram of the hypothesised relationships between the latent variables and their corresponding observed indicators.

imputed latent traits $\tilde{Y}_1$ and $\tilde{Y}_2$ with a latent variable measuring the Eurobarometer respondents propensity toward immigrants and derived from the Eurobarometer survey considering the following questions:

- Please tell me whether each of the following statements evokes a positive or negative feeling for you?
 - Immigration of people from other EU Member States (qb7_1)
 - Immigration of people from outside the EU (qb7_2)

measured on a 4-point Likert scale (1: "Very positive", 4: "Very negative") .

- For each of the following statements, please tell me whether you totally agree, tend to agree, tend to disagree or totally disagree.
 - Immigrants contribute a lot to (OUR COUNTRY) (qb8_1)
 - (OUR COUNTRY) should help refugees (qb8_2)

measured on a 4-point Likert scale (1: "Totally agree", 4: "Totally disagree") .

2.3.3 Common variables and their harmonisation

Several common variables were identified for the integration of data from ESS10 and Eurobarometer93.1. Table 2 shows the common variables. The integration requires careful harmonisation of those variables to ensure consistency and comparability. For all the common variables, cases with missing values are removed from the integration.

Common variables	Measurement level	Type
Country	nominal	observed
Age	numeric	observed
Gender	nominal	observed
Occupation	nominal	observed
Marital status	nominal	observed
Domicile	nominal	observed
Nationality - Citizenship	nominal	observed
Subjective income	ordinal	observed
Political Orientation	ordinal	observed
Attachment to country	ordinal	observed
Life Satisfaction	ordinal	observed
Satisfaction with national economy	ordinal	observed
Trust in Institutions	numeric	latent

Table 2: Common variables

Common countries

- ESS10: code `cntry`
- Eurobarometer93.1: code `tnscntry`

ESS10			Eurobarometer93.1		
Category	Code	Sample	Category	Code	Sample
Belgium	BE	1341	BE	3	1008
Bulgaria	BG	2718	BG	4	1057
Czechia	CZ	2476	CZ	9	1009
Estonia	EE	1542	EE	12	1006
Finland	FI	1577	FI	15	1028
France	FR	1977	FR	16	1006
Greece	GR	2799	EL	13	1016
Croatia	HR	1592	HR	17	1030
Hungary	HU	1849	HU	18	1059
Ireland	IE	1770	IE	19	1005
Italy	IT	2640	IT	21	1027
Lithuania	LT	1659	LT	23	1005
Netherlands	NL	1470	NL	29	1004
Portugal	PT	1838	PT	32	1056
Slovenia	SI	1252	SI	36	1012
Slovakia	SK	1418	SK	37	1077

Socio-demographic information

• Age:

	ESS10	Eurobarometer93.1
Question	How old were you at your last birthday?	Could you please tell me your exact age?
Variable Code	Age of respondent agea	AGE EXACT d11
Measurement level	numeric	numeric
Harmonisation	none	

• Gender:

	ESS10	Eurobarometer93.1
Question	Are you: Male, Female, or Do not identify with either of these categories?	Are you: Man or Woman?
Variable Code	gender	GENDER
Code	gndr	d10
Measurement level	nominal	nominal
Categories	male: 1 female: 2 None of the above/ Non binary/ do not recognize yourself in above categories: 3	Man: 1 Woman: 2
Harmonisation	Since the Eurobarometer does not include a non-binary option, responses in this category from the ESS are excluded	
	1 MAN	
	2 WOMAN	

• **Occupation:**

	ESS10	Eurobarometer93.1
Question	Which of these descriptions best describes your main activity in the last seven days?	What is your current occupation?
Variable Code Measurement level Categories	Main activity, last 7 days mnactic nominal Paid work 1 Education 2 Unemployed, looking for job 3 Unemployed, not looking for job 4 Permanently sick or disabled 5 Retired 6 Community or military service 7 Housework, looking after children, others 8 Other 9	OCCUPATION d15a nominal Responsible for ordinary shopping, etc. 1 Student 2 Unemployed, temporarily not working 3 Retired, unable to work 4 Farmer 5 Fisherman 6 Professional (lawyer, etc.) 7 Owner of a shop, craftsmen, etc. 8 Business proprietors, etc. 9 Employed professional (employed doctor, etc.) 10 General management, etc. 11 Middle management, etc. 12 Employed position, at desk 13 Employed position, travelling 14 Employed position, service job 15 Supervisor 16 Skilled manual worker 17 Unskilled manual worker, etc. 18
Harmonisation 1 EMPLOYED 2 UNEMPLOYED 3 RETIRED 4 IN EDUCATION	ESS 1,7 3,4,8 5,6 2	EUROBAROMETER 1,5,6,7,8,9,10,11,12,13,14,15,16,17,18 3 4 2

• **Marital status :**

	ESS10	Eurobarometer93.1
Question	What is your legal marital status?	What is your marital status?
Variable Code	Legal marital status maritalb	MARITAL STATUS d7
Measurement level	nominal	nominal
Categories	Legally married 1 In a legally registered civil union 2 Legally separated 3 Legally divorced/Civil union dissolved 4 Widowed/Civil partner died 5 None of these (NEVER married or in legally registered civil union) 6	(Re-)Married: without children 1 (Re-)Married: children this marriage 2 (Re-)Married: children prev marriage 3 (Re-)Married: children this/prev marriage 4 Single liv w partner: without children 5 Single liv w partner: childr this union 6 Single liv w partner: childr prev union 7 Single liv w partner: childr this/prev union 8 Single: without children 9 Single: with children 10 Divorced/Separated: without children 11 Divorced/Separated: with children 12 Widow: without children 13 Widow: with children 14
Harmonisation	ESS	EUROBAROMETER
1 LEGALLY MARRIED OR IN A CIVIL UNION OR LIVING WITH	1,2	1,2,3,4,5,6,7,8
2 SINGLE	6	9,10
3 DIVORCED OR SERATED	3,4	11,12
4 WIDOW	5	13,14

• **Domicile:**

	ESS10	Eurobarometer93.1
Question	Which phrase on this card best describes the area where you live?	Would you describe the place where you live as...?
Variable Code Measurement level Categories	Domicile domicil nominal A big city 1 Suburbs or outskirts of big city 2 Town or small city 3 Country village 4 Farm or home in countryside 5	TYPE OF COMMUNITY d25 nominal Rural area or village 1 Small/middle town 2 Large town 3
Harmonisation 1 A BIG CITY OR LARGE TOWN 2 SMALL MIDDLE TOWN 3 RURAL AREA OR VILLAGE	ESS 1,2 3 4,5	EUROBAROMETER 3 2 1

• **Nationality – Citizenship:**

	ESS10	Eurobarometer93.1
Question	Are you a citizen of [country]?	What is your nationality?
Variable Code Measurement level Categories	Citizen of country ctzcntr binary Yes 1 No 2	Countries in the sample from q1.1 to q1.36 binary for each country Not mentioned 0 Mentioned 1
Harmonisation 1 YES 0 NO	ESS 1 2	EUROBAROMETER 1 0

• **Subjective income:**

	ESS10	Eurobarometer93.1
Question	Which of the descriptions on this card comes closest to how you feel about your household's income nowadays?	During the last year, did you have difficulties paying your bills?
Variable Code Measurement level Categories	Feeling about household's income nowadays hincfel ordinal Living comfortably on present income 1 Coping on present income 2 Difficult on present income 3 Very difficult on present income 4	DIFFICULTIES PAYING BILLS - LAST YEAR d60 ordinal Most of the time 1 From time to time 2 Almost never/never 3
Harmonisation 1 YES 2 NO	ESS 1 2 3,4	EUROBAROMETER 3 2 1

• **Education:**

This variable is excluded due to irreconcilable differences in classification systems. The ESS uses a detailed international standard to categorize educational levels, while the Eurobarometer employs a simplified, nationally tailored system, rendering harmonisation impractical.

Attitudes and opinion

• Political Orientation :

Question	In politics, people sometimes talk of left and right. Where would you place yourself on this scale?	In political matters, people sometimes refer to 'left' and 'right.' Where would you place yourself on this scale?
Variable Code Measurement level Categories	Placement on left right scale lrscale ordinal from 0=left to 10= right	LEFT-RIGHT PLACEMENT d1 ordinal from 1=left to 10= right
Harmonisation 1 LEFT 2 CENTRE 3 RIGHT	Considering the distribution on the two surveys, the first two categories of the ESS variable are collapsed	

• Attachment to country:

Question	How emotionally attached do you feel to [your country]?	Please tell me how attached you feel to (OUR COUNTRY)
Variable Code Measurement level Categories	How emotionally attached to [country] atchctr ordinal from 0=Not at all emotionally attached to 10= Very emotionally attached	ATTACHMENT TO COUNTRY qc1a_2 ordinal from 1=Very attached to 4= Not at all attached
Harmonisation 1 NOT AT ALL ATTACHED 2 NOT VERY ATTACHED 3 FAIRLY ATTACHED 4 VERY ATTACHED	ESS 0,1,2 3,4,5 6,7,8 9,10	EUROBAROMETER 4 3 2 1

• **Life Satisfaction:**

	ESS10	Eurobarometer93.1
Question	All things considered, how satisfied are you with your life as a whole nowadays?	How satisfied are you with your life in general?
Variable	How satisfied with life as a whole	LIFE SATISFACTION
Code	stflife	d70
Measurement level	ordinal	ordinal
Categories	from 0=Extremely dissatisfied to 10=Extremely satisfied	from 1=Very satisfied to 4=Not at all satisfied
Harmonisation	ESS	EUROBAROMETER
1 NOT AT ALL SATISFIED	0,1,2	4
2 NOT VERY SATISFIED	3,4,5	3
3 FAIRLY SATISFIED	6,7,8	2
4 VERY SATISFIED	9,10	1

• **Satisfaction with national economy:**

	ESS10	Eurobarometer93.1
Question	On the whole how satisfied are you with the present state of the economy in [country]?	How would you judge the current situation in each of the situation of the (NATIONALITY) economy
Variable	How satisfied with present state of economy in country	SITUATION: NATIONAL ECONOMY
Code	stfeco	qa1a_2
Measurement level	ordinal	ordinal
Categories	from 0=Extremely dissatisfied to 10=Extremely satisfied	from 1=Very good to 4=Very badd
Harmonisation	ESS	EUROBAROMETER
1 NOT AT ALL SATISFIED	0,1,2	4
2 NOT VERY SATISFIED	3,4,5	3
3 FAIRLY SATISFIED	6,7,8	2
4 VERY SATISFIED	9,10	1

- **Trust in Institutions:** Trust in institution is measured through a latent trait. The following table provides details on the observed indicator and their rating scales.

	ESS10	Eurobarometer93.1
Question	Please tell me on a score of 0-10 how much you personally trust each of the institutions I read out.	How much trust you have in certain institutions. For each of the following institutions, please tell me if you tend to trust it or tend not to trust it.
Variable code	Country's parliament trstprl	NATIONAL PARLIAMENT textttqa6a_10
Variable code	Legal system trstlgl	JUSTICE / LEGAL SYSTEM a6a_3
Variable code	Police trstplc	POLICE qa6a_4
Variable code	Political parties trstprt	POLITICAL PARTIES qa6a_2
Variable code	European Parliament trstep	EUROPEAN UNION qa6a_11
Variable code	United Nations trstun	UNITED NATIONS qa6a_12
Measurement level	ordinal	binary
Categories	from 0=No trust at all to 10= Complete trust	1=Tend not to trust, 2=Tend to trust
Harmonisation 0 NO TRUST 1 TRUST	In order to apply the same IRT model to both surveys, the ESS variables have been dichotomized as follows: 0-5 = 2, and 6-10 = 1.	

2.3.4 Weighting schemes

ESS10

Different systems for weighting responses according to the sample design are provided in the ESS dataset. By default, it is recommended to always use analysis weight (`anweight`) for all analyses (Kaminska, 2020). This weight is suitable for all types of analysis, including studying a single country, comparing across countries, or analyzing groups of countries. For advanced users who may want to model a nonresponse correction themselves ESS provide a design weight (`dweight`).

- *analysis weight* - `anweight`: corrects for differential selection probabilities within each country as specified by sample design, for nonresponse, for noncoverage, and for sampling error related to the four post-stratification variables, and takes into account differences in population size across countries. It is constructed by first deriving the design weight, then applying a post-stratification adjustment, and then a population size adjustment.
- *design weight* - `dweight`: corrects only for differential selection probabilities. Kaminska (2020) does not recommend use of this weight without non-response correction.
- *post-stratified design weight* - `pspwght`: without the population size correction.

This weight can be used for single country analysis or for cross-country comparisons

Eurobarometer93.1

Information on sampling weights can be found here: [Weighting overview](#).

Up to six weights are available for the analysis of individual COUNTRIES resp. SAMPLES (East and West Germany, Great Britain and Northern Ireland) or NATIONS (United Germany and the United Kingdom).

Up to more than 20 European weights are provided for use in analyses of the European Union population as a whole or in accordance with its historical compositions in the process of enlargement or of the introduction of the common currency (EURO).

Given our interest in the separate analysis of countries, we will use the following weighting scheme:

- *WEIGHT RESULT FROM TARGET* (w_1): is a pure post-stratification weight which reproduces the real number of cases for each country. This weight assumes that all samples are analyzed separately.

3 Integrating ESS waves

Repeated cross-sectional surveys collect data over time from distinct, independently sampled subsets of the population. Econometricians have utilised repeated cross-sectional data to analyse dynamic changes via pseudo-panel models. These models categorise individuals into stable cohorts based on common attributes, such as birth year, then examine the average behaviour of these cohorts across survey waves. A different approach emphasises statistical matching. This method connects individuals across survey waves by aligning them according to common characteristics. In contrast to pseudo-panels, data fusion utilises respondent-level data, facilitating more accurate and adaptable analysis of evolving relationships across time. Statistical matching, in this context, facilitates the examination of population-level trends and individual-level variations in repeated cross-sectional surveys, serving as a powerful instrument for investigating dynamic attitudinal and behavioural transformations.

More details on the literature on statistical matching for repeated cross-sectional surveys are provided in Deliverable 2.1.

3.1 Second case study: Investigating the effect of the the COVID-19 pandemic on Trust in Institutions and Satisfaction with Country Conditions

As second case study, we plan to apply statistical matching methodology to two waves of the European Social Survey conducted before and after the COVID-19 pandemic to explore its impact on Trust in Institutions and Satisfaction with Country Conditions. By matching respondents across waves based on several socio-demographic variables, this approach will enable the construction of a pseudo-longitudinal dataset. This dataset will allow for the examination of dynamic changes over time while preserving the richness of respondent-level data. The study aims to identify shifts trust and satisfaction with national conditions and institutions.

In our analysis, we will initially consider the ESS Round 9 (2018, hereafter ESS9) and the ESS Round 11 (2023, hereafter ESS11).

We are planning to extend the analysis considering multiple waves.

3.1.1 Geographical coverage

- ESS9
 - Austria, Belgium, Bulgaria, Switzerland, Cyprus, Czech Republic, Germany, Denmark, Estonia, Spain, Finland, France, the United Kingdom, Croatia, Hungary, Ireland, Iceland, Italy, Lithuania, Latvia, Montenegro, the Netherlands, Norway, Poland, Portugal, Serbia, Sweden, Slovenia, and Slovakia.
- ESS11
 - Austria, Belgium, Bulgaria, Switzerland, Cyprus, Germany, Spain, Finland, France, the United Kingdom, Greece, Croatia, Hungary, Ireland, Israel, Iceland, Italy, Lithuania, Latvia, Montenegro, the Netherlands, Norway, Poland, Portugal, Serbia, Sweden, Slovenia, and Slovakia.

The common countries are 26: Austria, Belgium, Bulgaria, Switzerland, Cyprus, Germany, Spain, Finland, France, the United Kingdom, Croatia, Hungary, Ireland, Iceland, Italy, Lithuania, Latvia, Montenegro, the Netherlands, Norway, Poland, Portugal, Serbia, Sweden, Slovenia, and Slovakia.

3.1.2 Matching variables

To match units across the two surveys, we will select the matching variables from:

- the **socio-demographic profile** core module (ESS9: questions F1 – F61, ESS11: questions F1 – F61) which contain information on:
 - gender (*gendr*),
 - age (*agea*),
 - marital status (*marsts*),
 - type of area (*domicil*),
 - education (*eisced*),
 - occupation of respondent (*mnactic*),
 - feeling about household's income (*hincfel*)
 - satisfied with life as a whole (*stflife*)
 - happy (*happy*)
- some variables related **to political interest, religiosity and perceived discrimination**:
 - How interested in politics (*polintr*)
 - Voted last national election (*vote*)
 - Placement on left right scale (*lrscale*)
 - How emotionally attached to [country] (*atchctr*)

- How emotionally attached to Europe (atcherp)
- How religious are you (rlgdgr)
- Member of a group discriminated against in this country (dscrgrp)

We also consider the dimensions of the Human Value scale and some latent traits related to political participation and trust in others,

3.1.3 Target variables

To investigate the impact of the pandemic on Trust in Institutions and Satisfaction with Country Conditions, we will consider the following variables:

• Trust in Institutions

- trstprl - Trust in country's parliament
- trstlgl - Trust in the legal system
- trstplc - Trust in the police
- trstplt - Trust in politicians
- trstprt - Trust in political parties
- trstep - Trust in the European Parliament
- trstun - Trust in the United Nations

• Satisfaction with Country Conditions

- stfeco - How satisfied with present state of economy in country
- stfgov - How satisfied with the national government
- stfдем - How satisfied with the way democracy works in country
- stfedu - State of education in country nowadays
- stfhlth - State of health services in country nowadays

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